

Ankur Desai

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EDUCATION

Michigan State University

B.S. in Computer Science

Expected, May 2028

East Lansing, MI

- **GPA:** 3.5 / 4.0
- **Relevant Coursework:** Linear Algebra, Discrete Mathematics, Object-Oriented Software Design, Computer Organization and Architecture, Data Structures & Algorithms

EXPERIENCE

Claude Builder Club @ MSU

Vice President & Co-Founder

Oct. 2025 - Present

East Lansing, MI

- Automated end-to-end member provisioning pipeline (GitHub API + Slackbot SDK), reducing onboarding friction by 97% (15min → 30sec) and eliminating manual access control errors for 50+ members.
- Built QR code event check-in system with AWS S3 bucket caching and unique token generation for automated attendance tracking.
- Established CI/CD pipeline with GitHub Actions enforcing 90% coverage gates and 30+ unit & integrations tests; reduced PR merge time through parallelized test execution and branch protection policies.

MindMosaic

Software Engineer

Aug. 2025 – Nov. 2025

East Lansing, MI

- Developed full-stack note-taking application; integrated OAuth2.0 with JWT key rotation; implemented CI/CD pipeline maintaining 85% coverage across PyTest/Jest testing suites.
- Architected chapter mapping system enabling users to create bidirectional knowledge graphs, implementing polyglot persistence (PostgreSQL + Neo4j) handling recursive relationship queries faster than relational joins.
- Optimized graph query performance by 48% (850ms → 440ms) through Cypher query refactoring, database indexing on relationship types, and Alembic-managed schema migrations.

PROJECTS

Mira — AI Assistant | *Next.js, Electron, FastAPI, PostgreSQL, WebHook*

May 2025 – Nov. 2025

Distributed audio analysis platform with intelligent background processing and smart task execution

- Developed hybrid RAG retrieval system combining BM25 lexical search with Sentence-BERT semantic embeddings (384-dim), improving context retrieval accuracy by 35% over baseline keyword matching.
- Optimized FastAPI inference service with lazy-loaded DI pattern, reducing ML model cold-start latency by 86% (6,643ms → 930ms).
- Combined Electron desktop client with Next.js frontend, orchestrating real-time data flow via IPC channels and a dedicated webhook server to synchronize user actions and audio streams.

WizViz — SpartaHack X | *Python, MediaPipe, OpenCV, NumPy, TensorFlow*

Feb. 2025

Won Interactive Media Track with real-time computer vision game utilizing pose-estimation

- Designed low-latency gesture pipeline using MediaPipe Pose to map 33-point skeletal landmarks to game actions, achieving 60 FPS with sub-20ms input lag by optimizing the inference loop.
- Created custom vector-based physics engine with AABB collision detection to map 3D real-world movements into 2D game state in real-time.

TECHNICAL SKILLS

Languages Python, C++, Java, JavaScript/TypeScript, SQL, Bash
Backend Frameworks FastAPI, SQLAlchemy, Redis, Alembic, Node.js, PyTest
Frontend Development React, Next.js, Tailwind CSS, Electron, Vite, Jest
Cloud & Infrastructure AWS (EC2, RDS, S3), Docker, GitHub Actions, PostgreSQL, Neo4j